HEaaN

A Scalable Privacy-Preserving Machine Learning Using Homomorphic Encryption for Bioinformatician

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Homomorphic Encryption (HE) is a cryptographic scheme that enables arbitrary computations over encrypted data. It offers perfect protection for data at rest, in use, and in transit because encrypted data can be analyzed without decryption. We introduce HEaaN, a new scalable privacy-preserving data analysis tool that uses HE technology. HEaaN provides various statistics and machine learning toolkits, such as linear models, logistic regression, and more. The package offers user-friendly APIs similar to those of popular data analysis tools like Pandas and scikit-learn in Python and R. Additionally, HEaaN supports data analysis such as Polygenic Risk Scores and Ancestry Inference and integrates with tools developed within R/Bioconductor packages.

Homomorphic Encryption

- Homomorphic Encryption (HE) is a ground-breaking cryptographic protocol that allows computations to be conducted directly on encrypted data.
- This innovation provides unparalleled protection for data at all stages - whether it's at rest, in use, or in transit, ensuring analysis can be performed without ever needing to decrypt the data.
- The adoption of HE technology promises a new era of privacy-preserving data analysis.

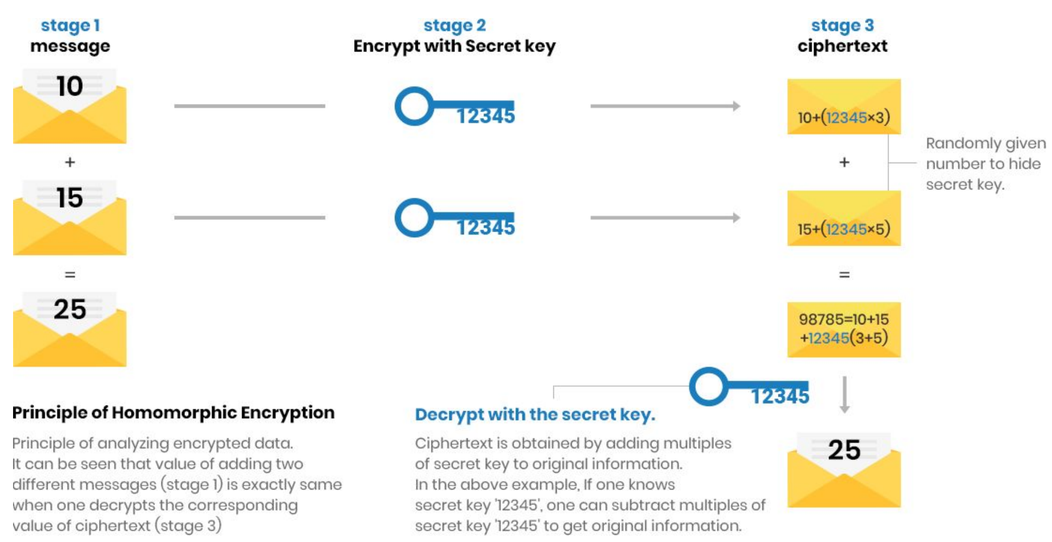

Figure from CryptoLab Tech Blog

Privacy-Preserving ML

- Privacy-preserving machine learning focuses on developing ML algorithms that can operate on encrypted data.
- This approach ensures the confidentiality and privacy of sensitive information during the learning process.
- Through mechanisms like homomorphic encryption, privacy-preserving machine learning allows us to glean valuable insights from data without compromising its security or privacy.

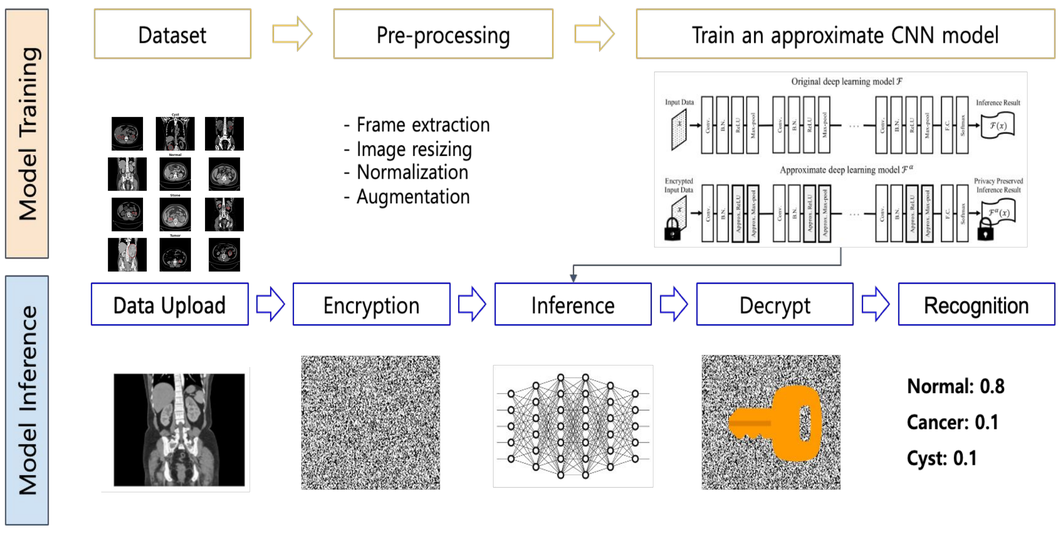

Figure from HEaaN-Kidney Tech Report (2023)

Usability for HEaaN

User-friendly APIs that are similar to those of famous data analysis tools such as tidyverse in R.

tidyverse (R)

```r
tb <- tibble(
    a = c(1L, 2L, 3L, 4L, 5L),
    b = c(-2, 0.3, 6, 11.3, 103.6),
    c = c("A", "B", "A", "A", "B")
)
tb$c <- as.factor(tb$c)
```

```r
tb <- tb %>%
  mutate("(a+b)" = tb$a + tb$b)
```

```r
mean <- mean(tb$a)
mean
```

```
[1] 3
```

```r
tb_f <- filter(tb, c == "A")
sd <- sd(tb_f$`(a+b)`)
sd
```

```
[1] 8.219692
```

```r
cov <- tb$a %>%
      cov(tb$b)
cov
```

```
[1] 55.55
```

HeaaN.stat (R)

```r
library(heaan.sdk.R)
import_heaan_sdk()
```

```r
hf <- HEFrame.from_frame(context, tb)
hf %>% encrypt()
```

```r
hf %>%
  mutate(hf["a"] + hf["b"])
```

```r
mean <- mean(hf["a"])
mean %>% decrypt() %>% to_series()
```

```
[1] 3
```

```r
sd <- hf["(a+b)"] %>%
    filter(hf["c"] == "A")
sd %>% sd() %>%
    decrypt() %>% to_series()
```

```
[1] 8.21972
```

```r
cov <- hf["a"] %>%
        cov(hf["b"])
cov %>% decrypt() %>% to_series()
```

```
[1] 55.55001
```

Supports: R (>= 3.6.0) https://hub.docker.com/r/cryptolabinc/heaan-stat

heaan.sdk.R — cryptolabinc/heaan-stat ☆ By cryptolabinc • Updated 18 days ago. Statistical analysis by homomorphic encryption. Image

R reticulate

HEaaN SDK (Python)

HEaaN Math (C++)

HEaaN Core (C++)

Docker

CPU (AVX512) | GPU (CUDA11)